

Week 14: Validated Playlist

Overview

This assignment turns Week 14 concepts into a small, finishable Java program or project milestone. Start from the provided template, preserve the public class filename, and run the program in IntelliJ before submitting.

Objectives

- Own an ArrayList inside an object.
- Validate text with Pattern and Matcher.
- Report invalid input with an exception.

Required reading - one new chapter

- [Think Java Chapter 13 - Objects of Arrays](#)

Optional supplemental reading - previously covered chapters

- [Chapter 5 - Conditionals and Logic](#)
- [Chapter 9 - Immutable Objects](#)

Interactive practice

- [Runestone: ArrayList coding practice](#)

Supplemental videos

- [ArrayLists](#)

Complete these steps

- Create Week14Playlist and a class named PlaylistDemo.
- Copy the starter into PlaylistDemo.java.
- Keep songs as an ArrayList field.
- Create a Pattern that accepts letters, numbers, spaces, and apostrophes in titles from 1 through 40 characters.
- In addSong, use matcher(title).matches().
- Throw IllegalArgumentException when a title does not match; otherwise add it.
- Use an enhanced for loop in showSongs.
- In main, add Here Comes the Sun, Happy, and Imagine.
- Use try/catch to demonstrate that a blank title is rejected without ending the program.
- Run and confirm the three songs and one validation message appear.
- Submit PlaylistDemo.java.

What will be checked

Use this list as your final review before submitting.

- Playlist owns an ArrayList.
- Pattern and matches are used to validate titles.

- Invalid input throws `IllegalArgumentException` and is caught in main.
- All three valid songs are printed.
- The submitted file contains no unfinished `TODO` marker from the starter template.